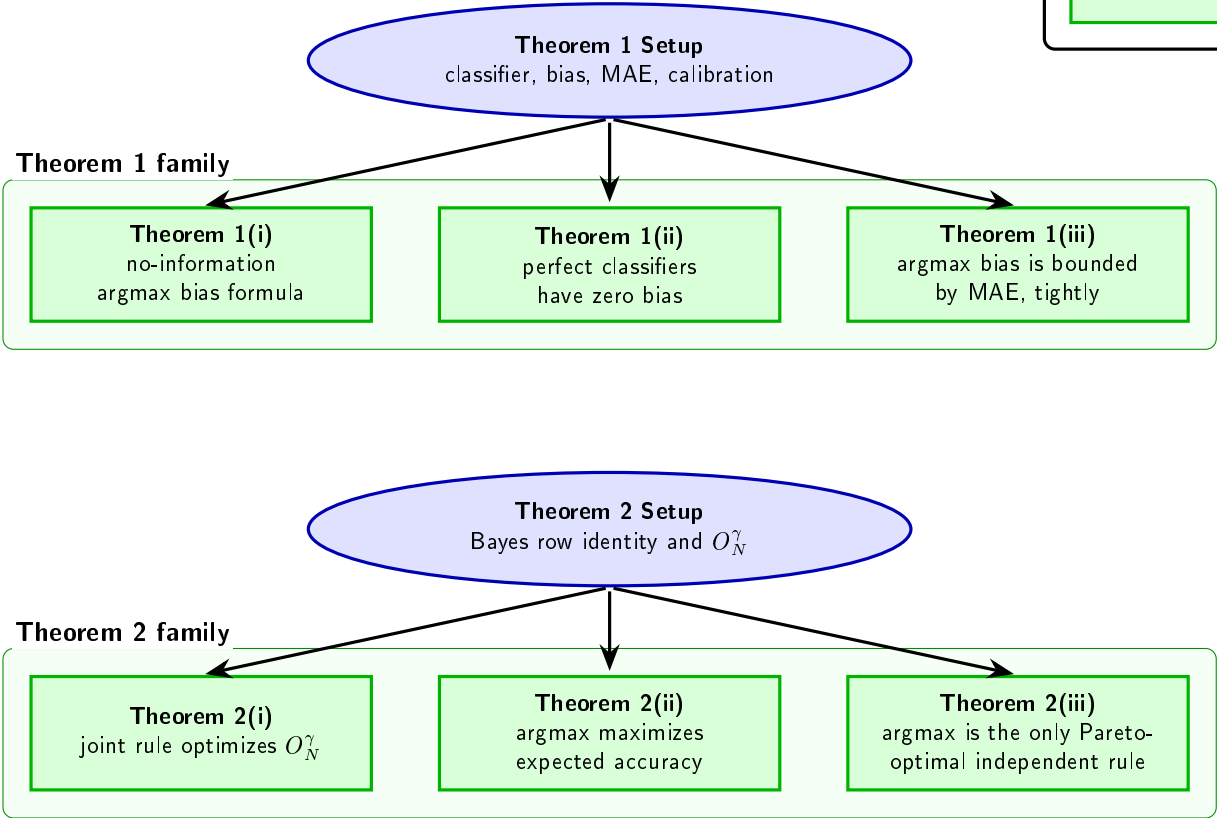
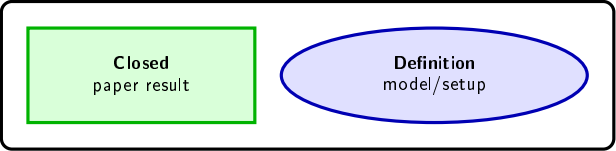


Legend



Audit note. Cached source text has only Theorem 1 and Theorem 2 as named results. Proof-route deviations and optional future strengthening are documented in the validation report; the green theorem boxes are closed paper-facing endpoints.